VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY

Deep Learning and Its Applications

Fashion-MNIST Image Classification

CO3133 · Assignment I
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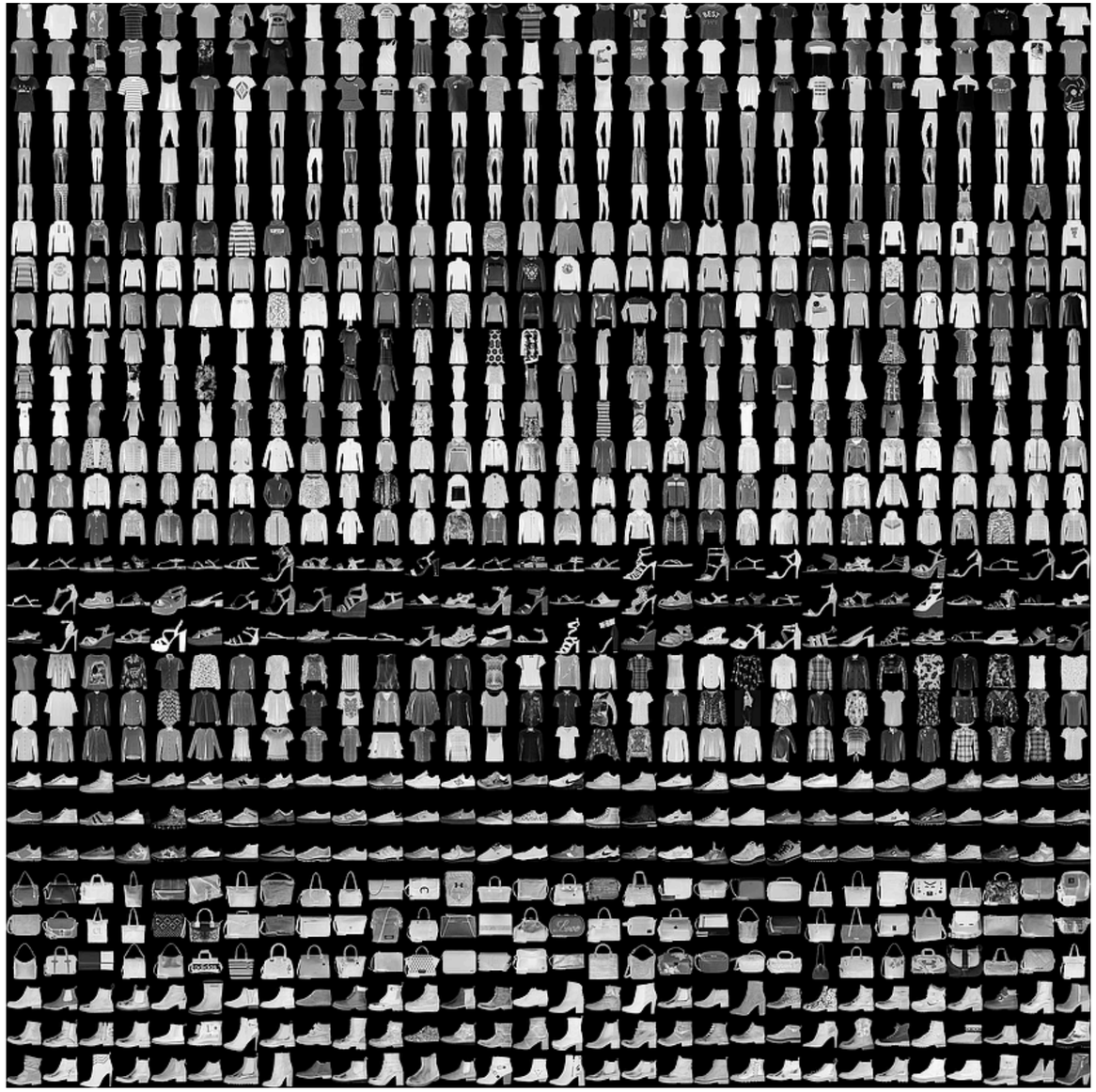


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1. Problem and Data

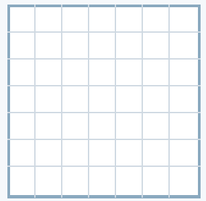
- Problem formulation
- Dataset Description
- Exploratory Data Analysis
- Data Splitting
- Data Preprocessing

Problem Formulation

Single-Label Multiclass Image Classification on Fashion-MNIST

INPUT

Input Image



28 × 28 Grayscale

$$x \in \mathbb{R}^{1\times 28\times 28}$$

→

INPUT REPRESENTATIONS

1D Vector	→	**Linear & MLP**
2D Spatial Grid	→	**CNN**
Row Sequences	→	**LSTM**
Image Patches	→	**ViT**

→

MODEL & SOFTMAX

Logits: $z = f_\theta(x) \in \mathbb{R}^{10}$

Softmax

$$p_{i,c} = \frac{\exp(z_{i,c})}{\sum_{k=0}^{9} \exp(z_{i,k})}$$

→

OUTPUT

Predicted Class

$$\hat{y} = \arg\max_{c\in\{0,\ldots,9\}} z_c$$

$$y \in \{0, 1, \ldots, 9\}$$

Research Objective: Evaluate how architectural inductive biases affect accuracy vs. compute efficiency on identical visual input.

Dataset Description: Fashion-MNIST

Drop-in Benchmark by Zalando Research (2017, MIT License)

70,000

Total Samples (60K Train / 10K Test)

10 Classes

100% Balanced (6,000 Train / 1,000 Test per class)

1 × 28 × 28

uint8 [0, 255] → float32 [0.0, 1.0]

85.8 MB / 188.2 MB

Raw disk / float32 RAM (60,000-image training pool)

Collection Pipeline

Catalog RGB Photos — **Auto Background Removal** — **Centering & Resizing** — **28 × 28 Grayscale**

Dataset Roles & 10 Labels

- MNIST (Digits): Unit sanity checks & pipeline debugging only.
- Fashion-MNIST: Benchmark for the MLP-CNN comparison.

10 Categories

0: T-shirt/top | 1: Trouser | 2: Pullover | 3: Dress | 4: Coat | 5: Sandal | 6: Shirt | 7: Sneaker | 8: Bag | 9: Ankle boot

Key Difficulties & Dataset Limitations

Intrinsic Bottlenecks in Low-Resolution Garment Recognition

Low-Resolution Ambiguity (28 × 28)

Fine details (buttons, collars, fabric weave) are lost.

High silhouette overlap in upper-body classes: T-shirt/top, Shirt, Pullover, Coat.

Representation Mismatch

- 1D Flattening: Destroys 2D spatial neighborhood structure.
- 1D Sequencing: Imposes artificial row/column ordering.

Commercial Catalog Bias

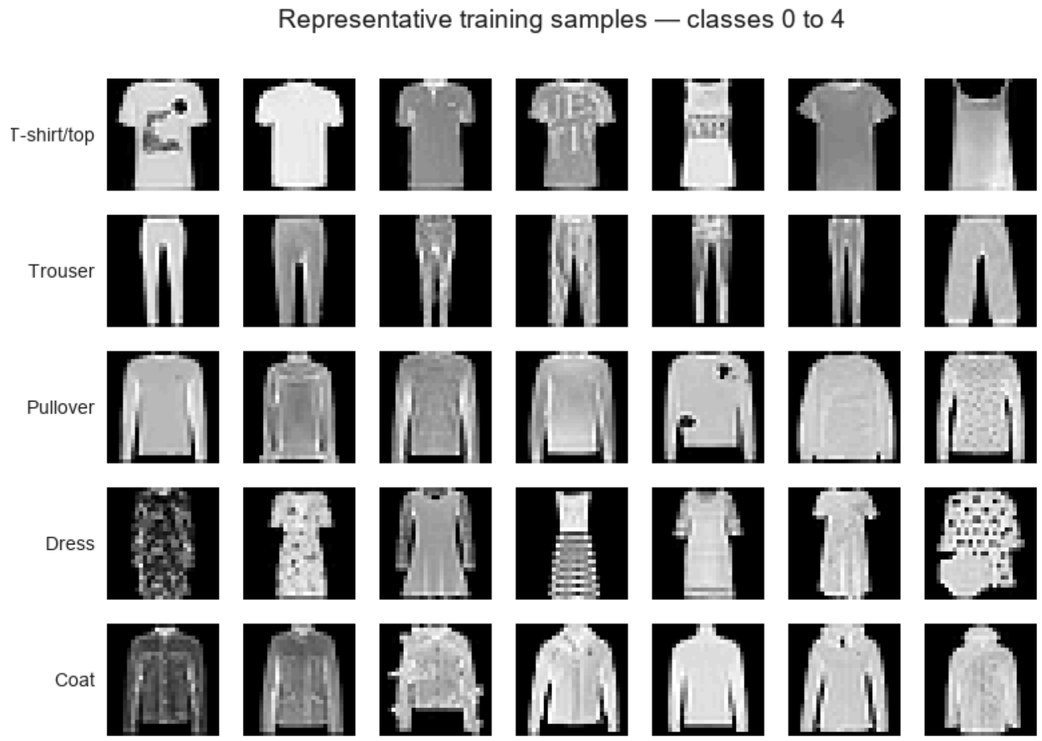
Studio lighting, centered framing, isolated black backgrounds.

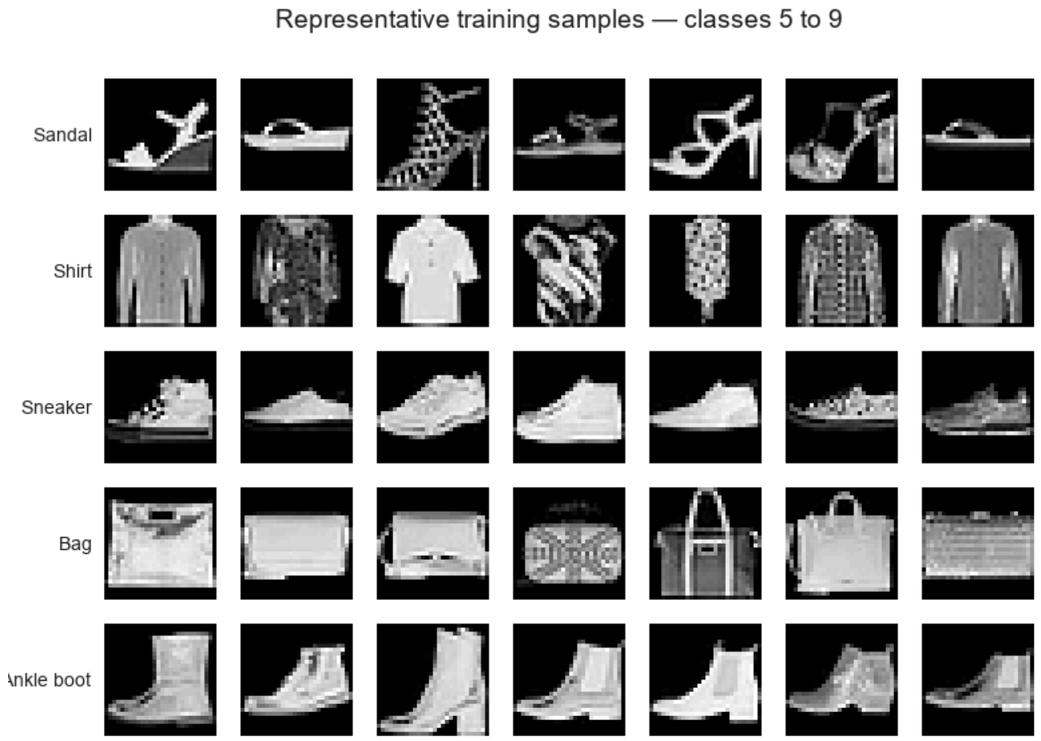
Does not directly generalize to cluttered in-the-wild consumer images.

Synthetic Balance & Reproducibility

- Uniform 10.0% class distribution eliminates class-starvation bias.
- Fixed split seed 42; training seeds 42, 43, 44.

EDA: Input and Representative Samples

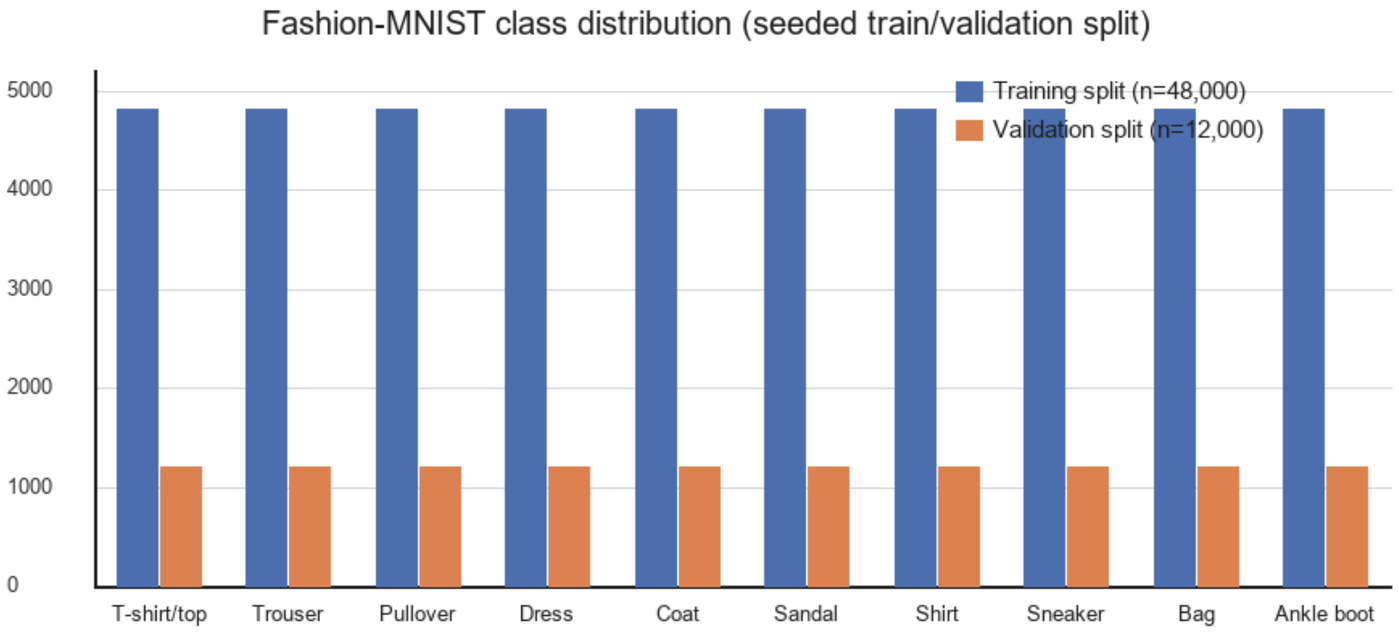




28 × 28 grayscale images, with 784 pixels and raw intensities in [0, 255]. All reported integrity checks passed.

Samples vary within each class. Upper garments have similar silhouettes at low resolution.

EDA: Class Distribution and Balance



Equal counts in all 10 classes

4,800 training images per class

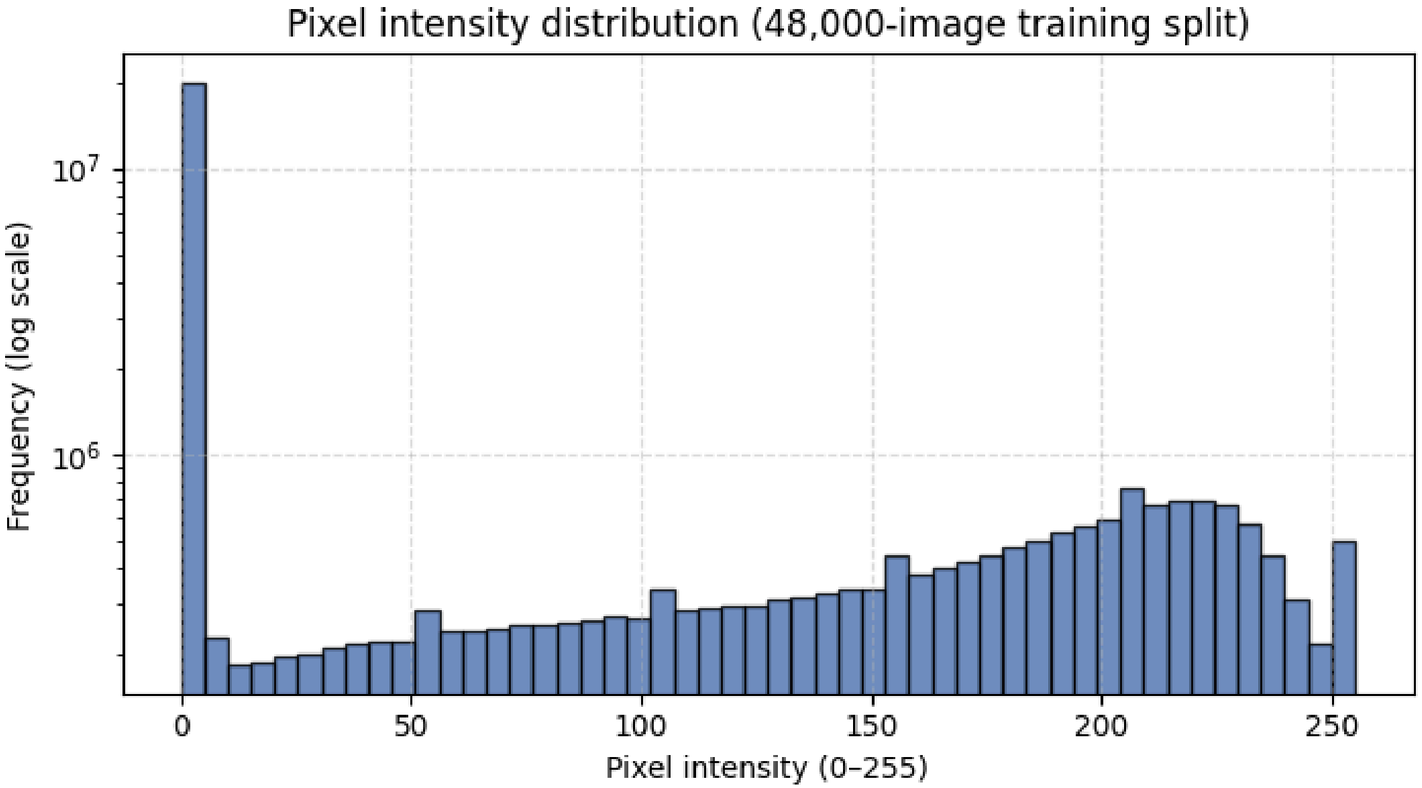
1,200 validation images per class

1,000 test images per class

Max/min class-count ratio: 1.0.
A constant-class predictor reaches 10% accuracy.

No class weighting or resampling.
Macro-F1 and per-class metrics still reveal differences in performance.

EDA: Pixel Intensities



Training pixels only

48,000 images
37,632,000 pixels
About 50.2% of pixels are zero.

Raw scale: μ = 72.99, σ = 90.06
After /255: μ = 0.2863, σ = 0.3532

Normalization: $x_{\text{norm}} = \dfrac{x_{\text{raw}}/255 - \mu_{\text{train}}}{\sigma_{\text{train}}}$

Data Splitting and Leakage Control

70,000 images

Fashion-MNIST



Stratified 80/20 split of the training pool, seed 42

Estimate μ and σ on unaugmented train images; augment train only

Preprocessing keeps validation and test data isolated

1 Split first

48k train
12k validation
10k official test

→

2 Augment train only

Rotate ±10°
Horizontal flip p = 0.5
Pad 2 px, then crop to 28 × 28

→

3 Convert

ToTensor
pixel range [0, 1]

→

4 Normalize

$(x - \mu_{train}) / \sigma_{train}$
$\mu = 0.2863$ • $\sigma = 0.3532$

TRAIN

Augmentation → tensor conversion → normalization

Random transforms change only training examples.

VALIDATION + TEST

No random augmentation

Same train-derived μ and σ; test remains held out.

Split before statistics • validation selects checkpoints • test is reserved for final evaluation.

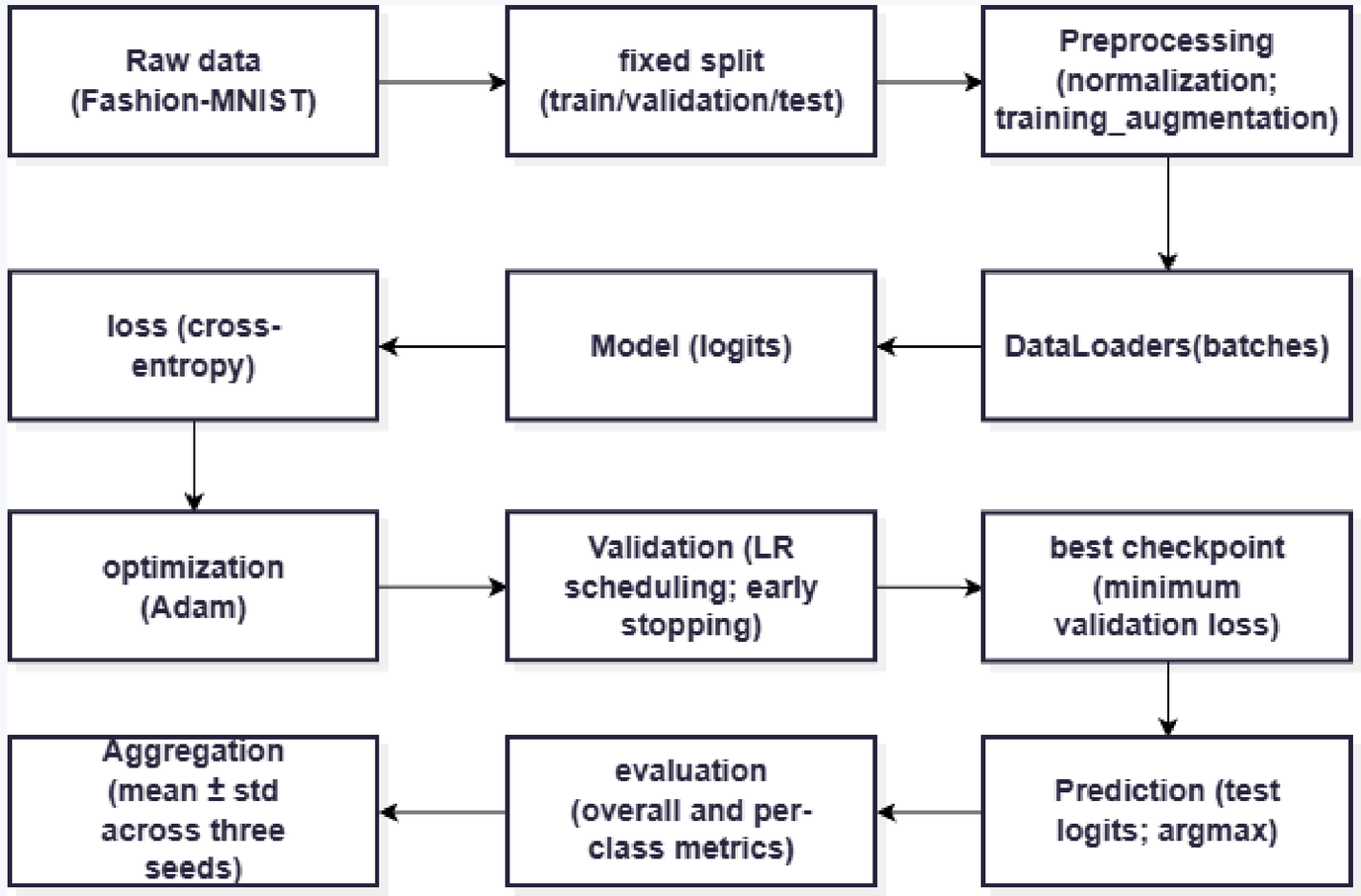
2. Main Methods

- Pipeline Overview
- Baseline (Linear Classifier)
- Multilayer Perceptron (MLP)
- Convolutional Neural Network (CNN)
- Long Short Term Memory (LSTM)
- Transformer

Pipeline Overview

Unified experimental pipeline and low-complexity reference model

UNIFIED EXPERIMENTAL PIPELINE



Baseline (Linear Classifier)

ARCHITECTURE & INPUT

Flattens 1 × 28 × 28 image to 1D vector

$x \in \mathbb{R}^{784}$

Single dense layer:

$z = W^T x + b (W \in \mathbb{R}^{784 \times 10}, b \in \mathbb{R}^{10}).$

No hidden layers, activations, or dropout

PARAMETERS & PROVENANCE

7,850

Trainable Parameters

(784 × 10) + 10

Self-implemented:

LinearClassifier class & flattening flow

Library:

torch.nn.Linear,
torch.nn.CrossEntropyLoss

STRENGTHS & LIMITATIONS

STRENGTHS

Simple architecture
Low parameter count
Interpretable weights

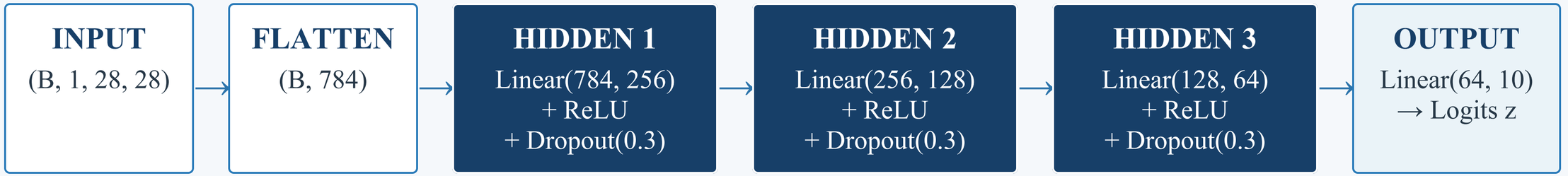
LIMITATIONS

Limited linear expressivity
No non-linear interactions
No 2D spatial hierarchy

Multilayer Perceptron (MLP) — Architecture & Forward Pass

Feed-forward architecture with explicit regularization and evaluation protocol

FORWARD PASS ARCHITECTURE



MATHEMATICAL FORMULATION

$$h^{(l)} = \text{Dropout}_{0.3}(\text{ReLU}(W^{(l)}h^{(l-1)} + b^{(l)})), l = 1, 2, 3.$$

$$z = W^{(4)}h^{(3)} + b^{(4)} \in \mathbb{R}^{10}$$

(raw unnormalized logits)

ACTIVATION & REGULARIZATION DESIGN

ReLU(u) = max(0, u) prevents linear-layer collapse.

Dropout(p = 0.3) active in training only (rescales by 1/(1 − p)); no BatchNorm used.

LOSS & EVALUATION PROTOCOL

Loss: **nn.CrossEntropyLoss** (internal log-softmax).

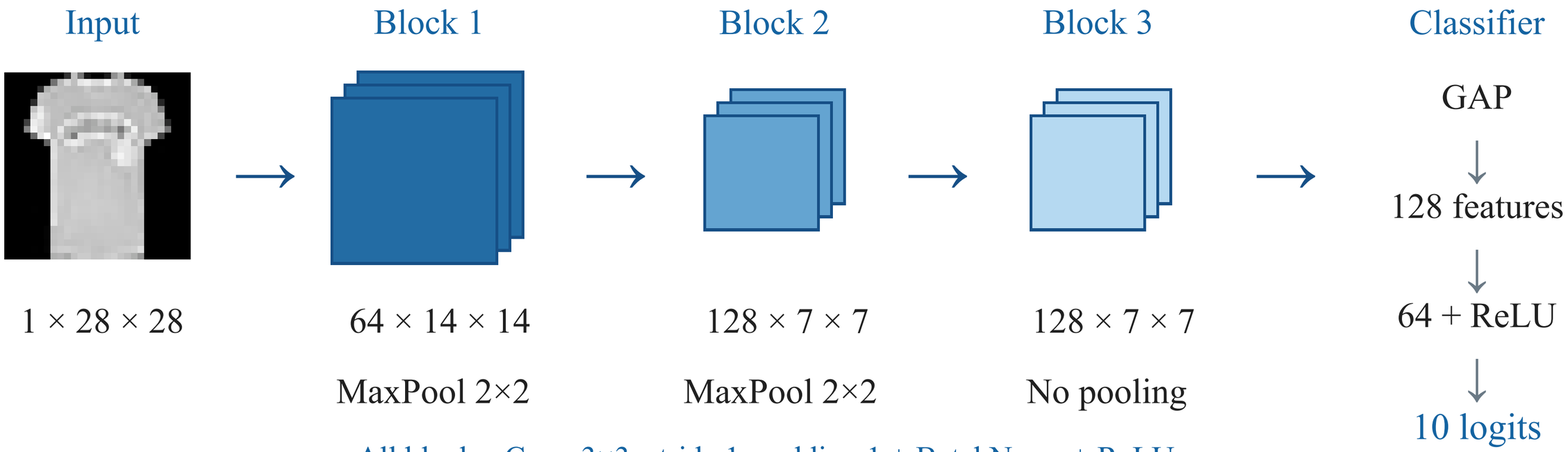
Primary: Top-1 accuracy.
Secondary: macro-F1, precision/recall, confusion matrix and inference time.

MLP — Parameter Budget Analysis

Block / Layer	Configuration	Parameter Formula	Count
Hidden 1	$\text{Linear}(784 \to 256)$	$(784 \times 256) + 256$	200,960
Hidden 2	$\text{Linear}(256 \to 128)$	$(256 \times 128) + 128$	32,896
Hidden 3	$\text{Linear}(128 \to 64)$	$(128 \times 64) + 64$	8,256
Output	$\text{Linear}(64 \to 10)$	$(64 \times 10) + 10$	650
Total Trainable			**242,762**

CNN: Architecture and Input

Custom 3-block CNN (SimpleCNN in the report)



All blocks: Conv 3×3, stride 1, padding 1 + BatchNorm + ReLU

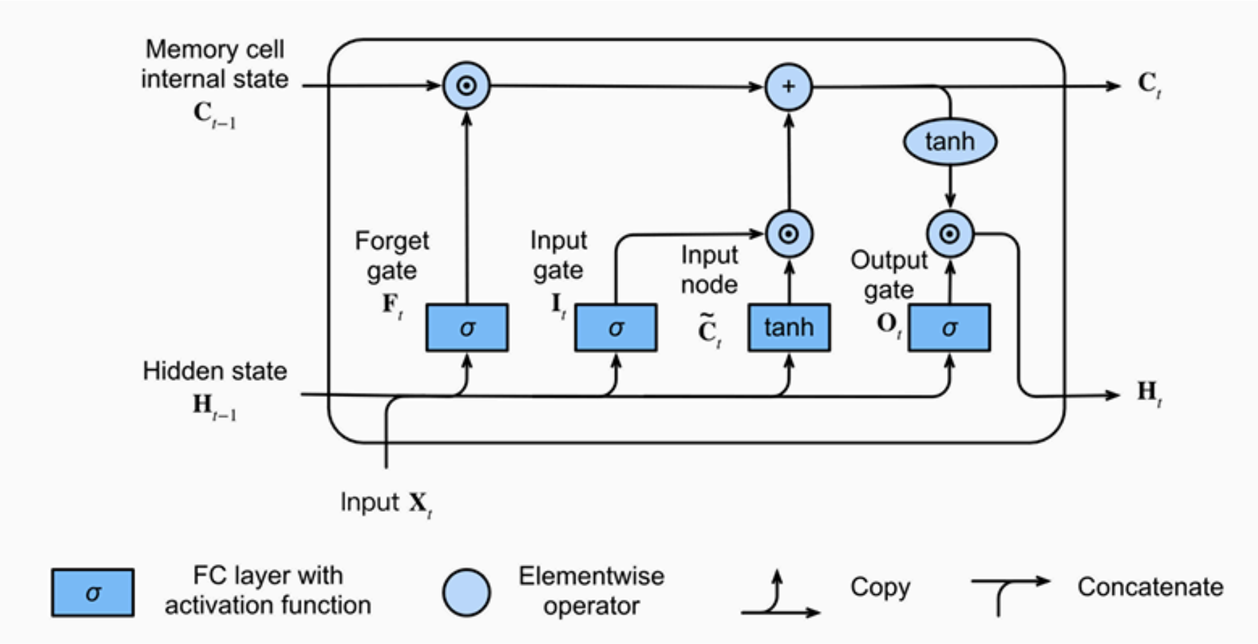
Blocks: Dropout2d 0.25 / 0.25 / 0.30
Classifier: Dropout 0.30

Input scaling: /255, then normalize with training $\mu = 0.2863$ and $\sigma = 0.3532$

CNN — Parameter Budget Analysis

Block / Layer	Configuration	Parameter Formula	Count
Block 1	$\text{Conv2d}(1 \to 64, 3 \times 3, \text{pad} = 1)$	$(64 \times 1 \times 3 \times 3) + 64$	640
	$\text{BatchNorm2d}(64)$	$\gamma(64) + \beta(64)$	128
Block 2	$\text{Conv2d}(64 \to 128, 3 \times 3, \text{pad} = 1)$	$(128 \times 64 \times 3 \times 3) + 128$	73,856
	$\text{BatchNorm2d}(128)$	$\gamma(128) + \beta(128)$	256
Block 3	$\text{Conv2d}(128 \to 128, 3 \times 3, \text{pad} = 1)$	$(128 \times 128 \times 3 \times 3) + 128$	147,584
	$\text{BatchNorm2d}(128)$	$\gamma(128) + \beta(128)$	256
Classifier	$\text{Linear}(128 \to 64)$	$(64 \times 128) + 64$	8,256
	$\text{Linear}(64 \to 10)$	$(10 \times 64) + 10$	650
Total Trainable			**231,626**

LSTM: Architecture and Input



$$\mathbf{S}_t = [\mathbf{H}_{t-1} \parallel \mathbf{X}_t] \in \mathbb{R}^{B \times (132+d)}$$

$$\mathbf{I}_t = \sigma\left(\mathbf{S}_t \mathbf{W}_i + \mathbf{1}_B \mathbf{b}_i^T\right),$$

$$\mathbf{F}_t = \sigma\left(\mathbf{S}_t \mathbf{W}_f + \mathbf{1}_B \mathbf{b}_f^T\right),$$

$$\mathbf{O}_t = \sigma\left(\mathbf{S}_t \mathbf{W}_o + \mathbf{1}_B \mathbf{b}_o^T\right),$$

$$\mathbf{G}_t = \tanh\left(\mathbf{S}_t \mathbf{W}_g + \mathbf{1}_B \mathbf{b}_g^T\right),$$

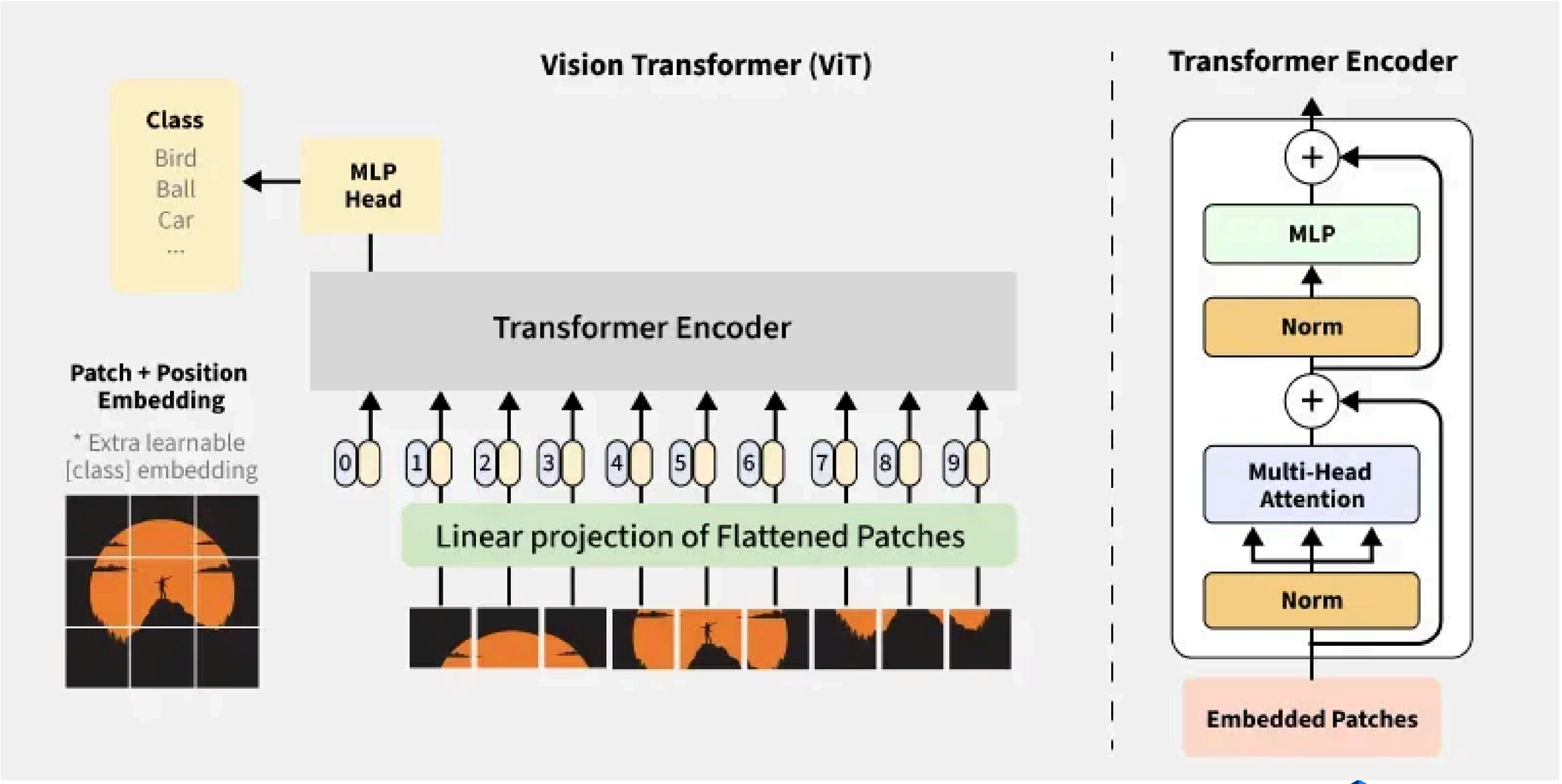
$$\mathbf{C}_t = \mathbf{F}_t \odot \mathbf{C}_{t-1} + \mathbf{I}_t \odot \mathbf{G}_t,$$

$$\mathbf{H}_t = \mathbf{O}_t \odot \tanh(\mathbf{C}_t).$$

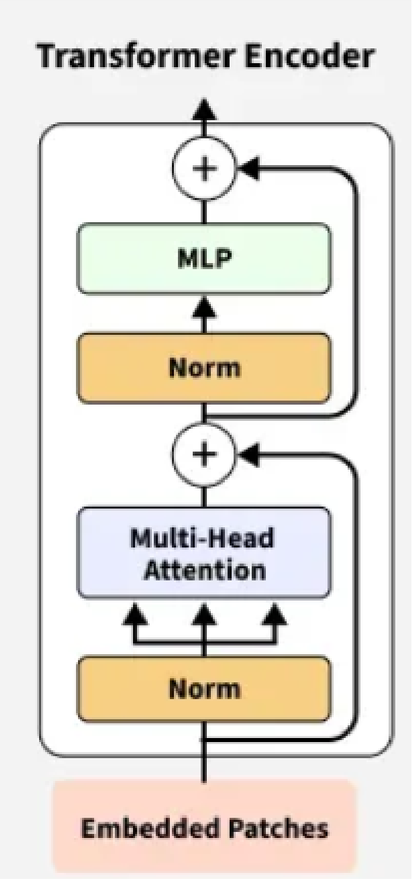
LSTM — Parameter Budget Analysis

Block / Layer	Configuration	Parameter Formula	Count
Recurrent	LSTM layer 1$(28 \to 132)$	$4 \cdot 132 \cdot (28 + 132) + 2 \cdot 4 \cdot 132$	85,536
	LSTM layer 2$(132 \to 132)$	$4 \cdot 132 \cdot (132 + 132) + 2 \cdot 4 \cdot 132$	140,448
Classifier	Linear$(132 \to 64)$	$(64 \times 132) + 64$	8,512
	Linear$(64 \to 10)$	$(10 \times 64) + 10$	650
Total Trainable			**235,146**

Transformer: Architecture and Input



Transformer: Architecture and Input



$$\mathbf{Z} \in \mathbb{R}^{B\times 50\times 64}$$

$$\mathbf{U}_{\text{attn}} = \text{LN}_1(\mathbf{Z}), \qquad \mathbf{U}_{\text{ffn}} = \text{LN}_2(\mathbf{Z}').$$

$$\mathbf{Q}_h = \mathbf{U}_{\text{attn}}\mathbf{W}_h^Q + \mathbf{b}_h^Q,$$
$$\mathbf{K}_h = \mathbf{U}_{\text{attn}}\mathbf{W}_h^K + \mathbf{b}_h^K,$$
$$\mathbf{V}_h = \mathbf{U}_{\text{attn}}\mathbf{W}_h^V + \mathbf{b}_h^V,$$

$$\mathbf{A}_h = \text{softmax}\left(\frac{\mathbf{Q}_h\mathbf{K}_h^T}{\sqrt{16}}\right),$$
$$\mathbf{H}_h = \text{Dropout}_{0.1}(\mathbf{A}_h)\mathbf{V}_h.$$

$$\text{MHA}(\mathbf{U}_{\text{attn}}) = \text{Concat}(\mathbf{H}_1, \mathbf{H}_2, \mathbf{H}_3, \mathbf{H}_4)\,\mathbf{W}^O + \mathbf{b}^O$$

$$\text{FFN}(\mathbf{U}_{\text{ffn}}) = \text{Dropout}_{0.1}(\text{GELU}(\mathbf{U}_{\text{ffn}}\mathbf{W}_1 + \mathbf{b}_1))\,\mathbf{W}_2 + \mathbf{b}_2,$$

$$\mathbf{Z}' = \mathbf{Z} + \text{Dropout}_{0.1}(\text{MHA}(\mathbf{U}_{\text{attn}})),$$
$$\mathbf{Z}'' = \mathbf{Z}' + \text{Dropout}_{0.1}(\text{FFN}(\mathbf{U}_{\text{ffn}})).$$

$$\mathbf{L} = \text{LN}_{\text{final}}\left(\mathbf{Z}^{(6)}_{[:,0,:]}\right)\mathbf{W}_{\text{cls}} + \mathbf{b}_{\text{cls}}$$

Transformer: Parameter Budget

Block / Layer	Configuration	Parameter Formula	Count
Tokenization	Linear$(16 \to 64)$ (patch projection)	$(64 \times 16) + 64$	1,088
	[CLS] token	1×64	64
	Positional embedding	50×64	3,200
Encoder (per layer)	Multi-head attention (Q, K, V, output proj.)	$(3 \cdot 64 \cdot 64 + 192) + (64 \cdot 64 + 64)$	16,640
	Linear$(64 \to 160)$	$(160 \times 64) + 160$	10,400
	Linear$(160 \to 64)$	$(64 \times 160) + 64$	10,304
	$2 \times$ LayerNorm(64)	$2 \times (64 + 64)$	256
	Subtotal per layer		37,600
Encoder (6 layers)		$6 \times 37{,}600$	225,600
Classifier	LayerNorm(64)	$64 + 64$	128
	Linear$(64 \to 10)$	$(10 \times 64) + 10$	650
Total Trainable			**230,730**

3. Experimental Setup

- Shared Training Setup
- Experimental Comparison

Shared training setup

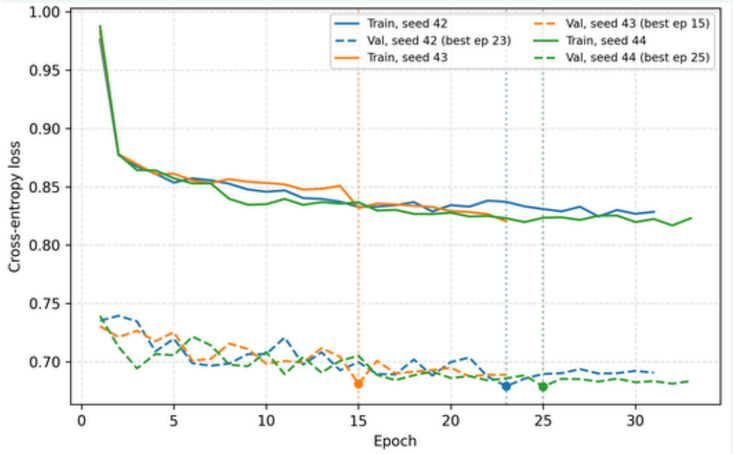
Optimizer	Adam($\beta1=0.9$, $\beta2=0.999$); cross-entropy loss; no weight decay
Learning rate	Initial 1×10^{-3}; minimum 1×10^{-6}
LR scheduler	ReduceLROnPlateau: factor 0.5, patience 8; absolute threshold 10^{-4} on validation loss
Early stopping	8 epochs without validation-loss improvement
Checkpoint	Lowest validation loss
Training seeds	42, 43, 44; fixed split seed 42
Hardware / precision	NVIDIA Tesla T4 GPU, Google Colab(Linux, Python 3.13.15, PyTorch 2.11.0+cu130, Torchvision 0.26.0+cu130, numpy 2.1.3), Precision: FP32
Batch size / epoch budget	128 train, 256 validation/test; maximum 200 epochs

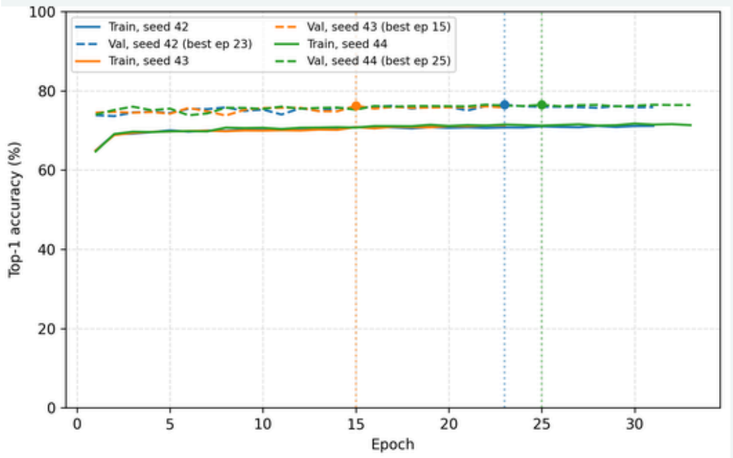
4. Main Results

- Training Behavior
- Quantitative results
- Qualitative results
- Error Analysis

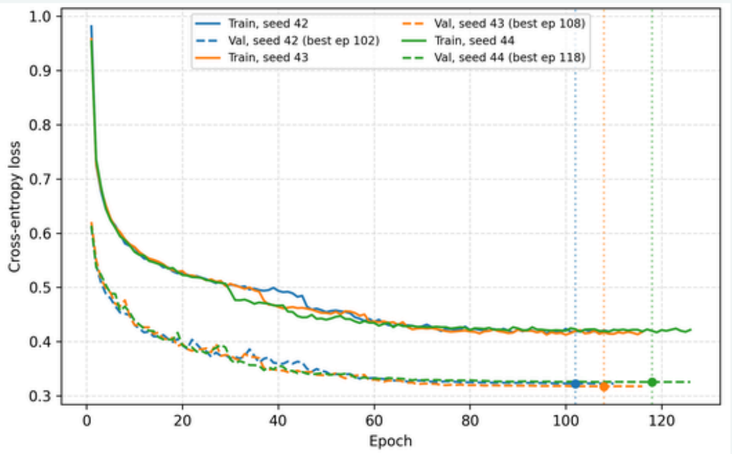
Training behavior: Linear, MLP & CNN

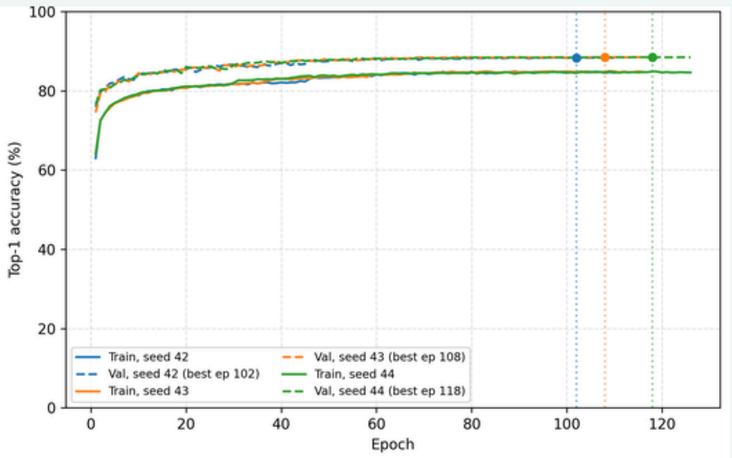
Linear



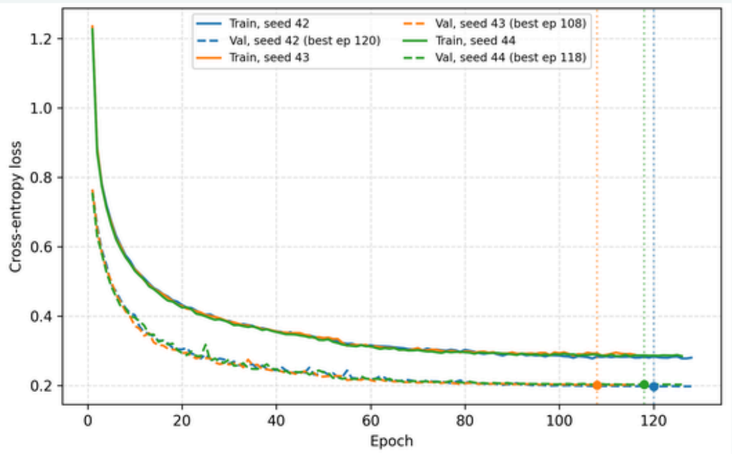


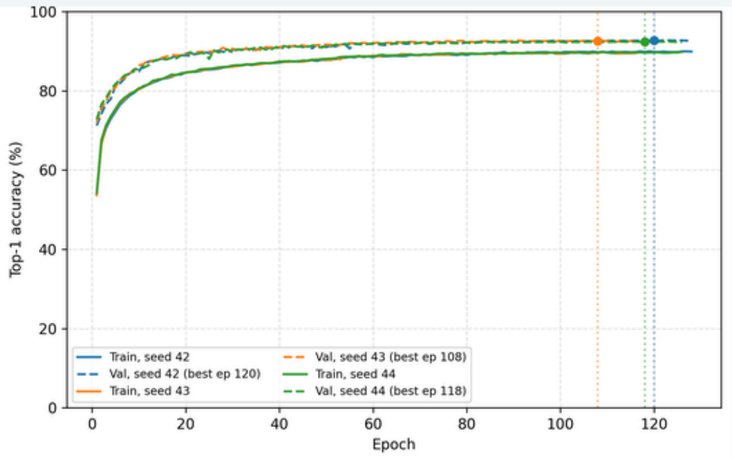
MLP





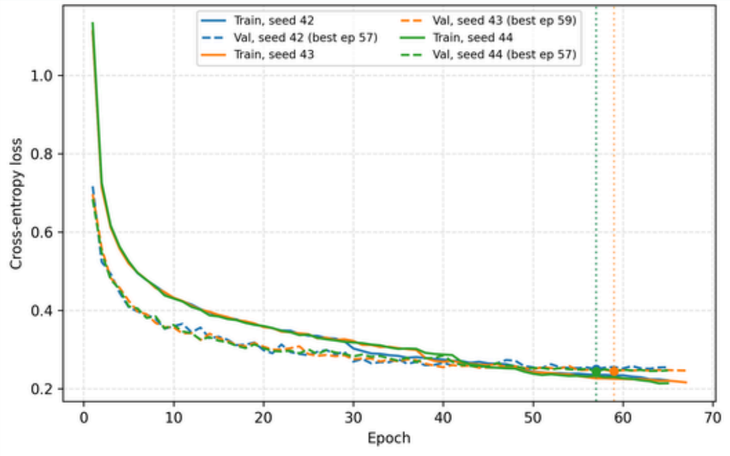
CNN

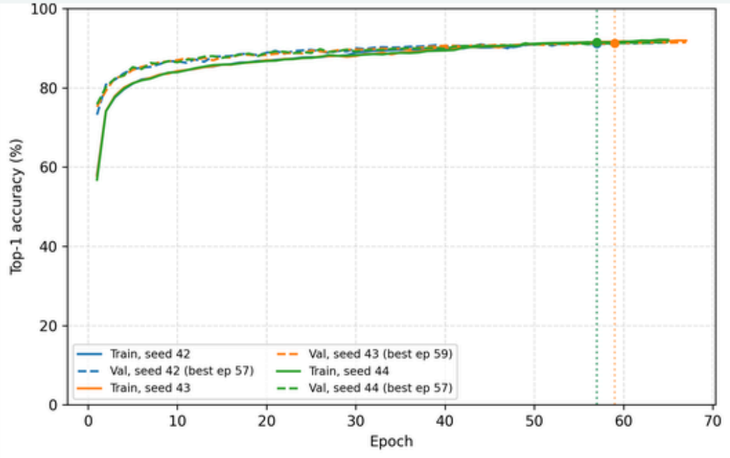




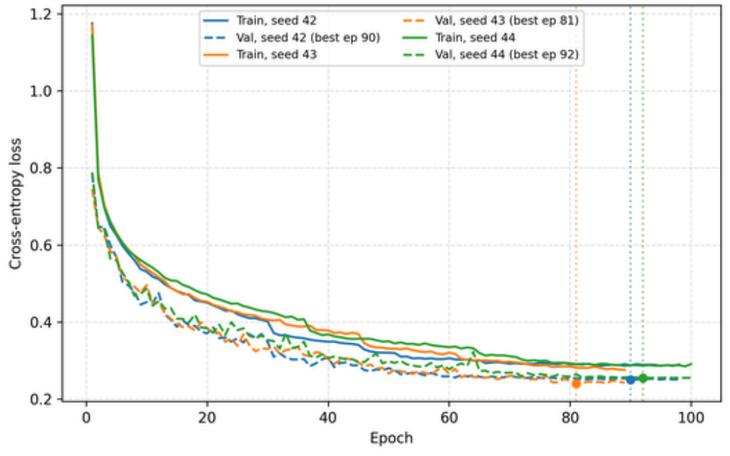
Training behavior: LSTM & Transformer

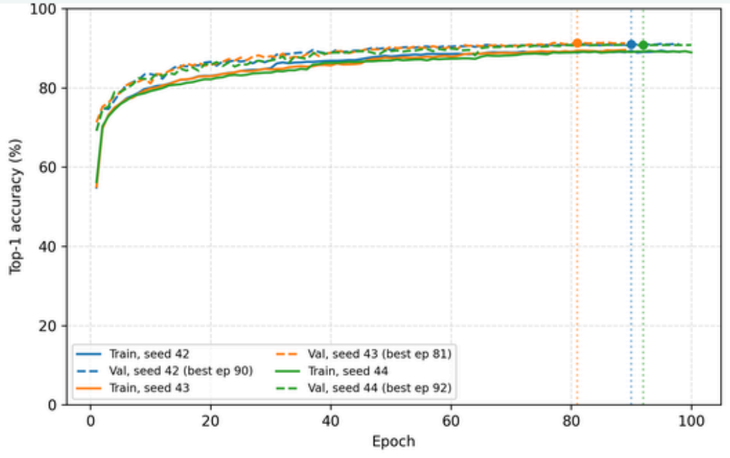
LSTM





Transformer





Quantitative results across five models

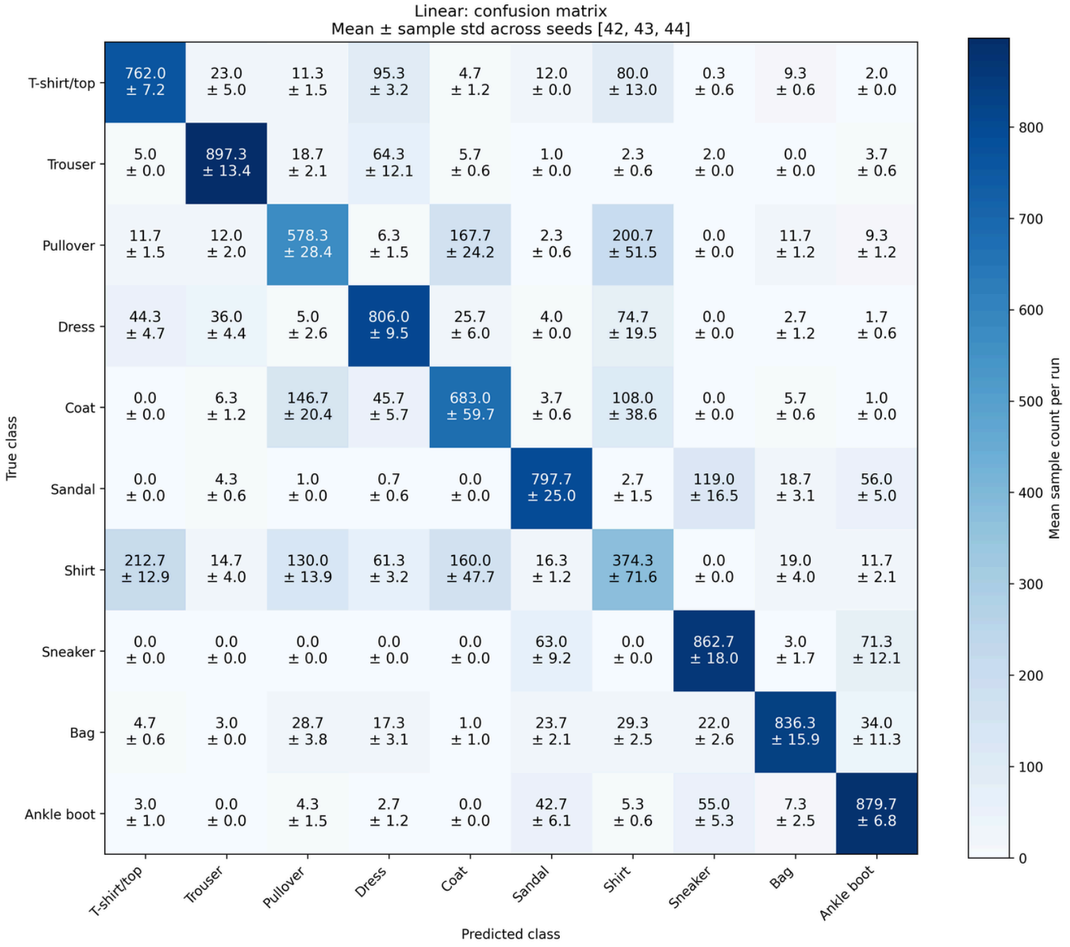
10,000 held-out test images • mean ± sample SD across training seeds 42, 43, 44

Model	Top-1 accuracy (%)	Macro-precision (%)	Macro-recall (%)	Macro-F1 (%)	Test CE loss
Linear	74.77 ± 0.35	74.49 ± 0.14	74.77 ± 0.35	74.48 ± 0.12	0.7141 ± 0.0028
MLP	87.30 ± 0.11	87.15 ± 0.10	87.30 ± 0.11	87.18 ± 0.10	0.3486 ± 0.0047
CNN	**91.54 ± 0.20**	**91.52 ± 0.19**	**91.54 ± 0.20**	**91.51 ± 0.19**	**0.2301 ± 0.0030**
LSTM	90.85 ± 0.10	90.79 ± 0.08	90.85 ± 0.10	90.80 ± 0.09	0.2654 ± 0.0032
Transformer	90.43 ± 0.22	90.43 ± 0.18	90.43 ± 0.22	90.37 ± 0.25	0.2643 ± 0.0094

Computational cost across five models

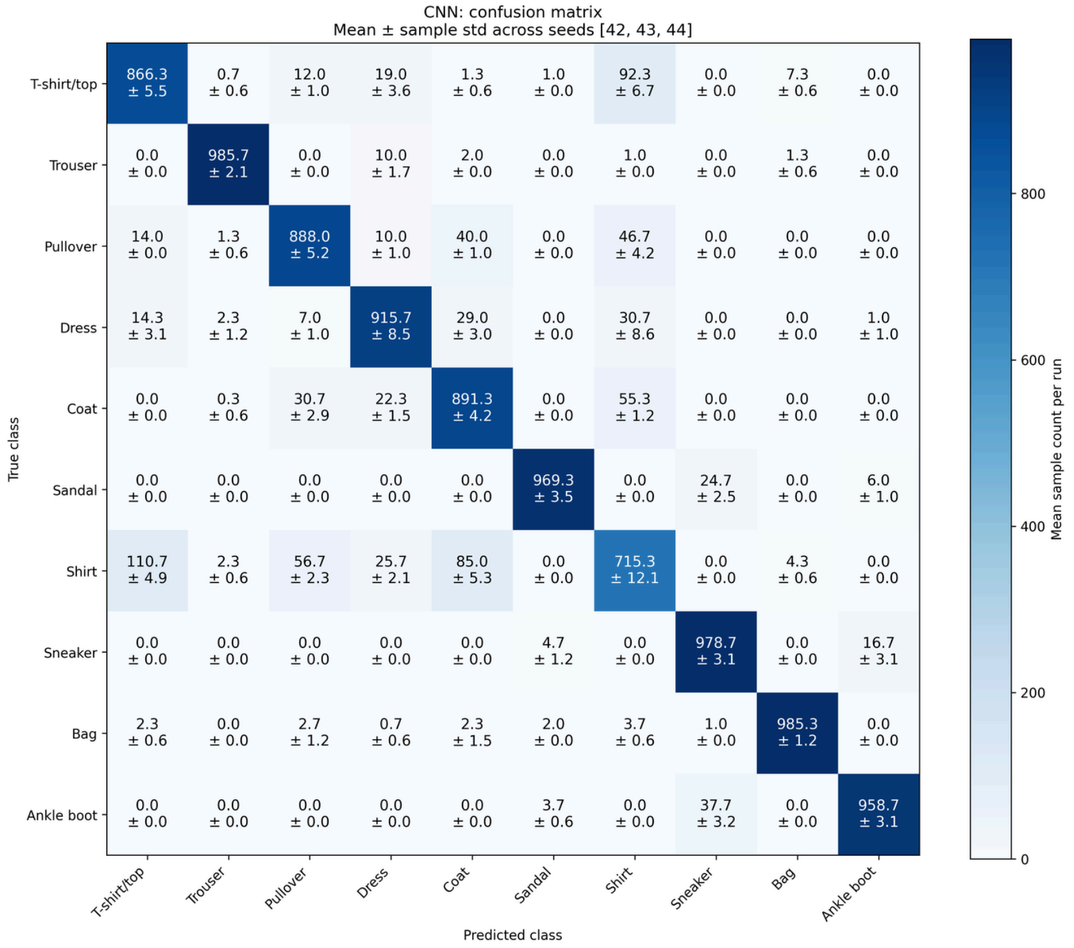
Model	Parameters	Epochs	Train time/run (s)	Train time/epoch (s)	Inference (ms/sample)
Linear	7,850	29.00 ± 5.29	526.7 ± 107.6	18.11 ± 0.51	**0.1681 ± 0.0062**
MLP	242,762	117.33 ± 8.08	2,141.9 ± 181.8	18.25 ± 0.80	0.1688 ± 0.0125
CNN	231,626	**123.33 ± 6.43**	**2,380.5 ± 213.7**	19.28 ± 0.77	0.1727 ± 0.0010
LSTM	235,146	65.67 ± 1.15	1,270.6 ± 26.6	19.36 ± 0.74	0.1804 ± 0.0047
Transformer	230,730	95.67 ± 5.86	2,183.0 ± 123.0	**22.83 ± 0.33**	0.1738 ± 0.0052

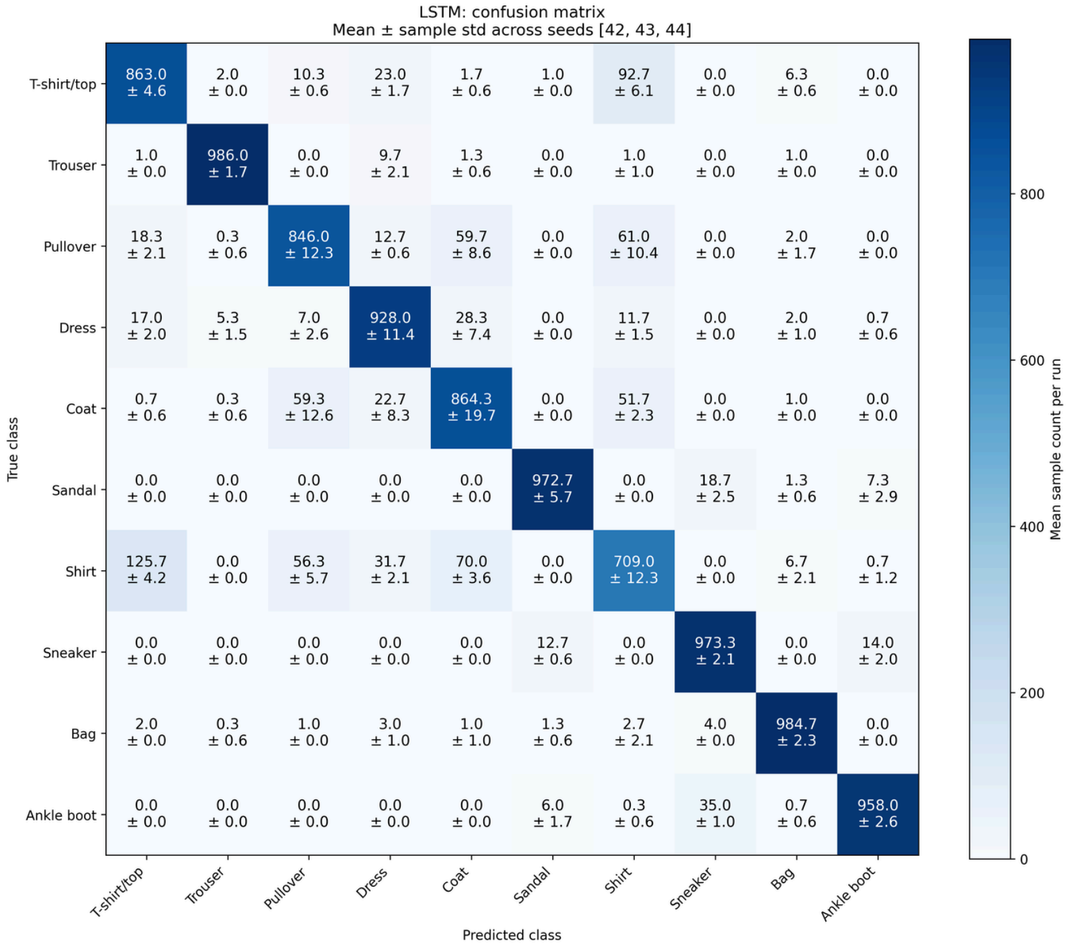
Confusion matrices: Linear and MLP



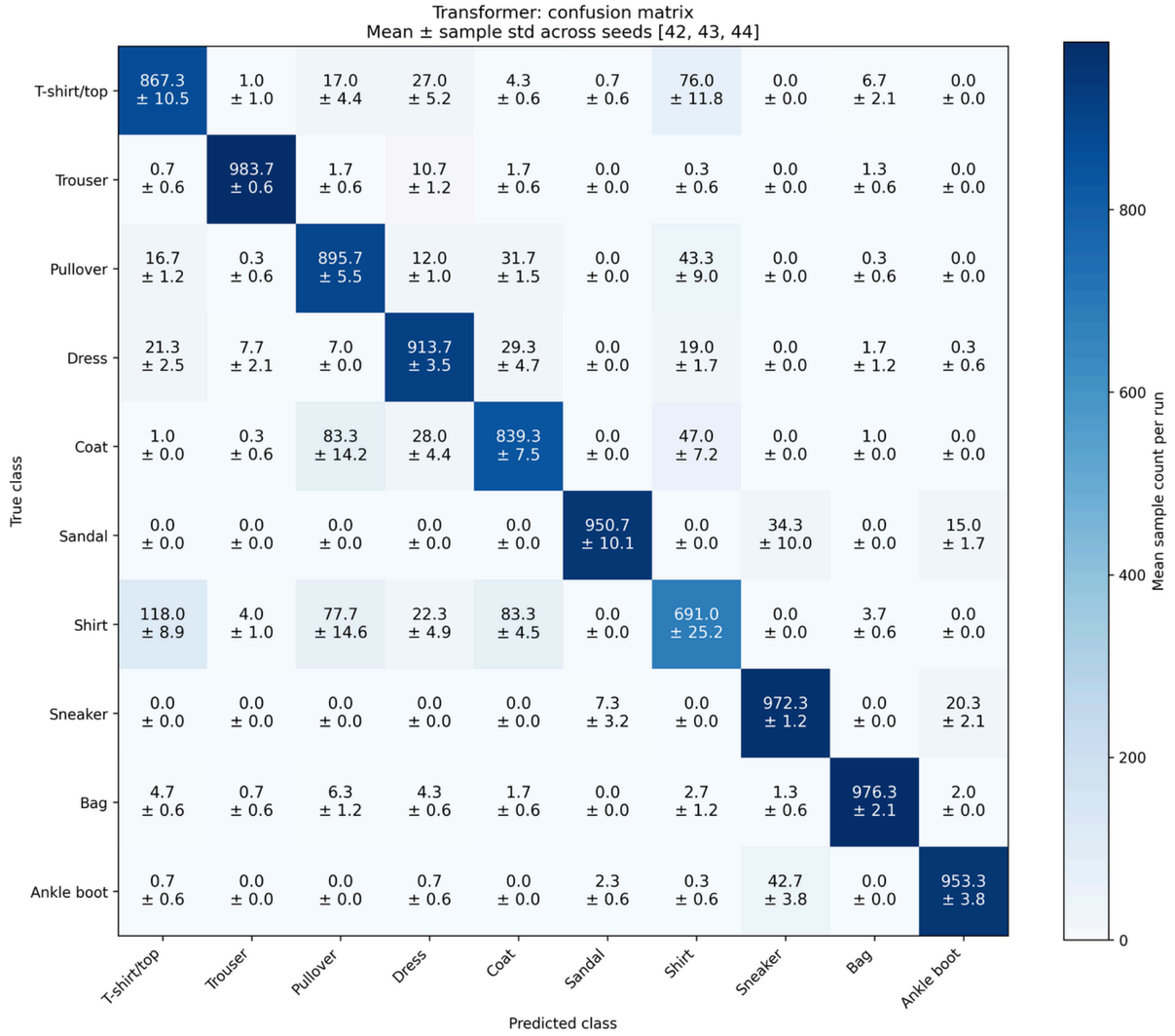


Confusion matrices: CNN and LSTM



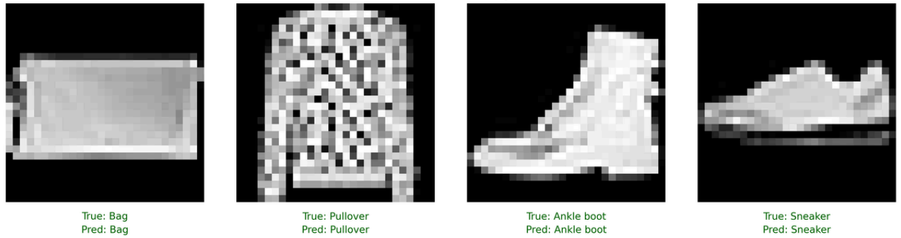


Confusion matrix: Transformer



Qualitative results · Linear baseline

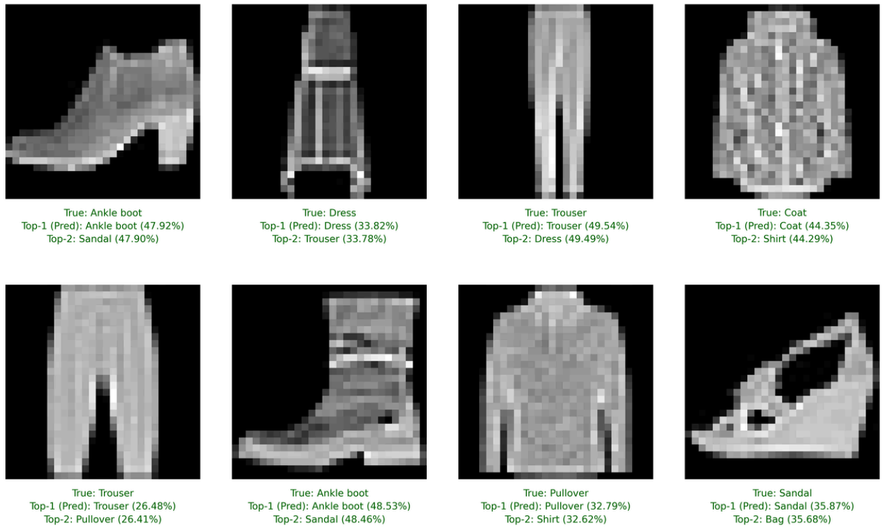
Linear / seed 42: Correct predictions



Linear / seed 42: Incorrect predictions

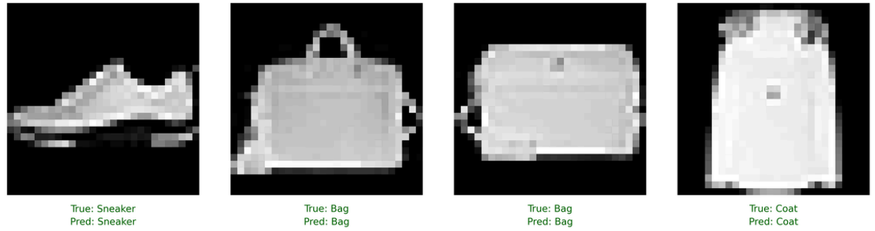


Linear / seed 42: Difficult but correct candidates



Qualitative results · MLP

MLP / seed 42: Correct predictions



True: Sneaker
Pred: Sneaker

True: Bag
Pred: Bag

True: Bag
Pred: Bag

True: Coat
Pred: Coat

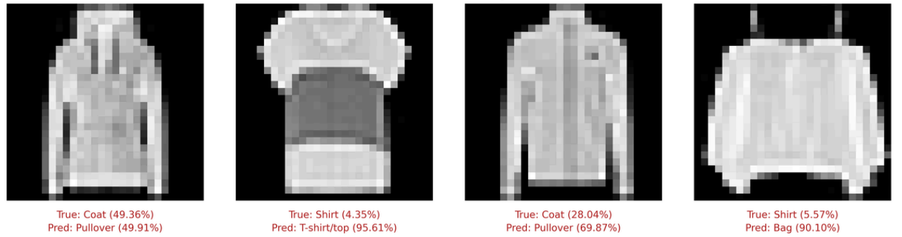
True: Sneaker
Pred: Sneaker

True: Sandal
Pred: Sandal

True: Coat
Pred: Coat

True: Coat
Pred: Coat

MLP / seed 42: Incorrect predictions



True: Coat (49.36%)
Pred: Pullover (49.91%)

True: Shirt (4.35%)
Pred: T-shirt/top (95.61%)

True: Coat (28.04%)
Pred: Pullover (69.87%)

True: Shirt (5.57%)
Pred: Bag (90.10%)

True: Coat (28.88%)
Pred: Pullover (69.40%)

True: T-shirt/top (24.46%)
Pred: Dress (45.78%)

True: Sandal (11.71%)
Pred: Sneaker (82.78%)

True: Coat (11.25%)
Pred: Shirt (75.61%)

MLP / seed 42: Difficult but correct candidates



True: Shirt
Top-1 (Pred): Shirt (46.84%)
Top-2: T-shirt/top (46.78%)

True: Coat
Top-1 (Pred): Coat (38.69%)
Top-2: Pullover (38.63%)

True: Shirt
Top-1 (Pred): Shirt (48.41%)
Top-2: T-shirt/top (48.34%)

True: Sneaker
Top-1 (Pred): Sneaker (49.93%)
Top-2: Ankle boot (49.83%)

True: Dress
Top-1 (Pred): Dress (47.95%)
Top-2: Coat (47.81%)

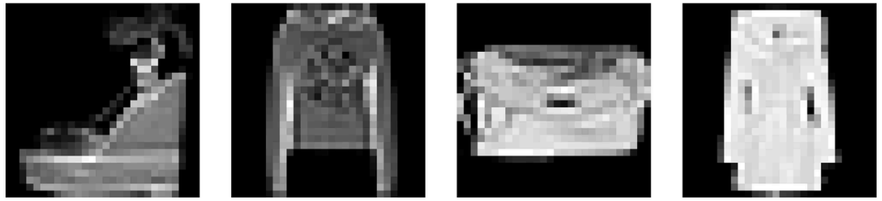
True: Shirt
Top-1 (Pred): Shirt (33.31%)
Top-2: Pullover (33.12%)

True: Shirt
Top-1 (Pred): Shirt (45.59%)
Top-2: T-shirt/top (45.23%)

True: T-shirt/top
Top-1 (Pred): T-shirt/top (50.16%)
Top-2: Shirt (49.79%)

Qualitative results · CNN

CNN / seed 42: Correct predictions



True: Sandal
Pred: Sandal

True: Pullover
Pred: Pullover

True: Bag
Pred: Bag

True: Coat
Pred: Coat

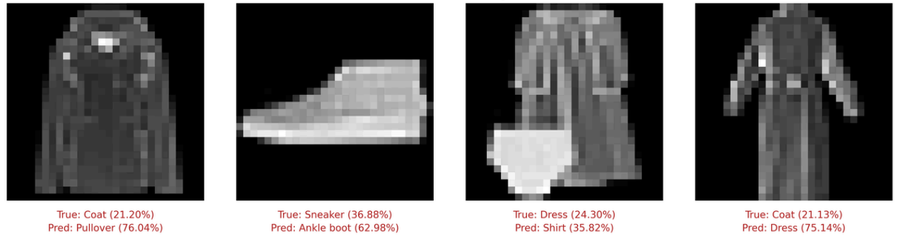
True: Coat
Pred: Coat

True: Trouser
Pred: Trouser

True: Coat
Pred: Coat

True: Dress
Pred: Dress

CNN / seed 42: Incorrect predictions



True: Coat (21.20%)
Pred: Pullover (76.04%)

True: Sneaker (36.88%)
Pred: Ankle boot (62.98%)

True: Dress (24.30%)
Pred: Shirt (35.82%)

True: Coat (21.13%)
Pred: Dress (75.14%)

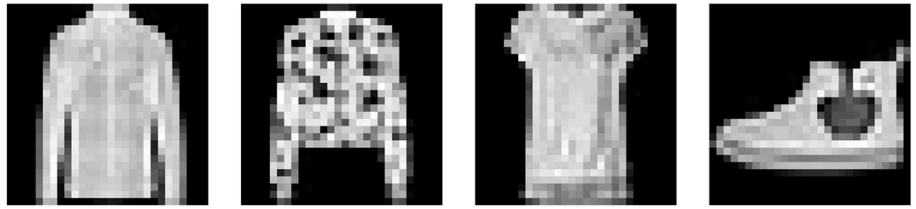
True: Shirt (45.67%)
Pred: Coat (53.11%)

True: Shirt (39.02%)
Pred: Coat (57.33%)

True: Coat (24.40%)
Pred: Dress (56.15%)

True: Ankle boot (24.91%)
Pred: Sneaker (74.90%)

CNN / seed 42: Difficult but correct candidates



True: Shirt
Top-1 (Pred): Shirt (49.32%)
Top-2: Coat (49.25%)

True: Coat
Top-1 (Pred): Coat (47.79%)
Top-2: Shirt (47.71%)

True: Shirt
Top-1 (Pred): Shirt (48.77%)
Top-2: T-shirt/top (48.68%)

True: Ankle boot
Top-1 (Pred): Ankle boot (49.41%)
Top-2: Sneaker (49.21%)

True: Coat
Top-1 (Pred): Coat (47.29%)
Top-2: Shirt (47.00%)

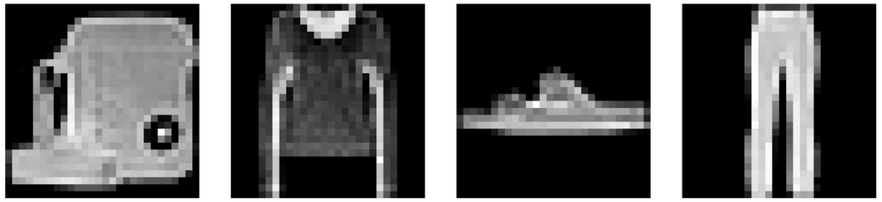
True: Dress
Top-1 (Pred): Dress (41.69%)
Top-2: Shirt (41.34%)

True: Dress
Top-1 (Pred): Dress (46.70%)
Top-2: Coat (46.27%)

True: Sneaker
Top-1 (Pred): Sneaker (50.23%)
Top-2: Ankle boot (49.63%)

Qualitative results · LSTM

LSTM / seed 42: Correct predictions



True: Bag
Pred: Bag

True: Pullover
Pred: Pullover

True: Sandal
Pred: Sandal

True: Trouser
Pred: Trouser

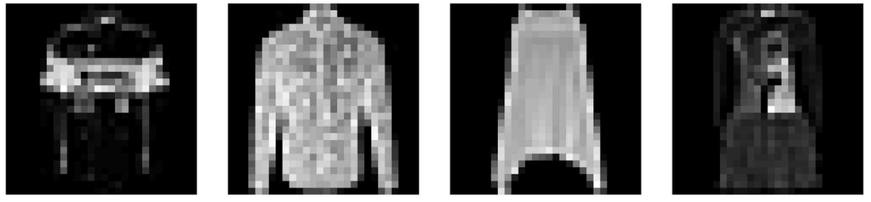
True: Dress
Pred: Dress

True: Sandal
Pred: Sandal

True: Dress
Pred: Dress

True: Coat
Pred: Coat

LSTM / seed 42: Incorrect predictions



True: Coat (2.16%)
Pred: Bag (48.60%)

True: Coat (4.34%)
Pred: Shirt (94.39%)

True: T-shirt/top (11.53%)
Pred: Shirt (69.52%)

True: Dress (33.25%)
Pred: T-shirt/top (42.15%)

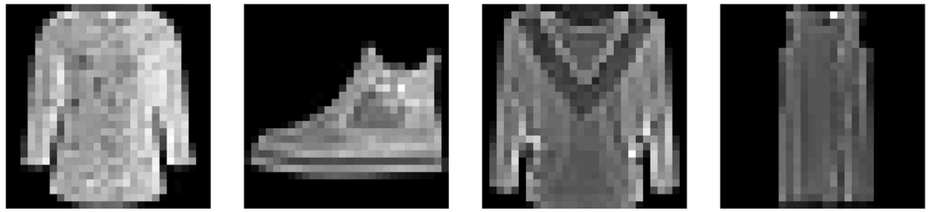
True: Pullover (24.59%)
Pred: Coat (75.36%)

True: Shirt (4.79%)
Pred: T-shirt/top (95.17%)

True: Shirt (1.61%)
Pred: T-shirt/top (95.74%)

True: Pullover (0.38%)
Pred: Coat (99.31%)

LSTM / seed 42: Difficult but correct candidates



True: T-shirt/top
Top-1 (Pred): T-shirt/top (36.13%)
Top-2: Shirt (36.06%)

True: Ankle boot
Top-1 (Pred): Ankle boot (50.00%)
Top-2: Sneaker (49.91%)

True: Pullover
Top-1 (Pred): Pullover (42.77%)
Top-2: Shirt (42.64%)

True: Dress
Top-1 (Pred): Dress (47.45%)
Top-2: T-shirt/top (47.24%)

True: T-shirt/top
Top-1 (Pred): T-shirt/top (47.82%)
Top-2: Dress (47.48%)

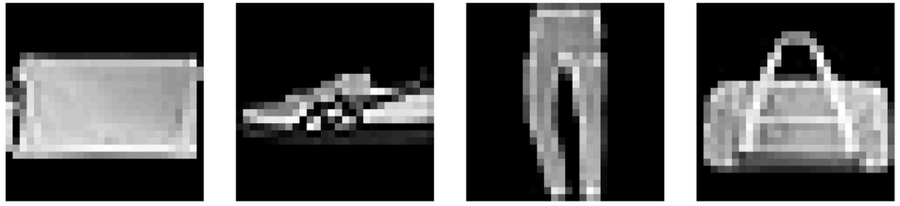
True: Sneaker
Top-1 (Pred): Sneaker (50.13%)
Top-2: Ankle boot (49.66%)

True: T-shirt/top
Top-1 (Pred): T-shirt/top (50.19%)
Top-2: Shirt (49.63%)

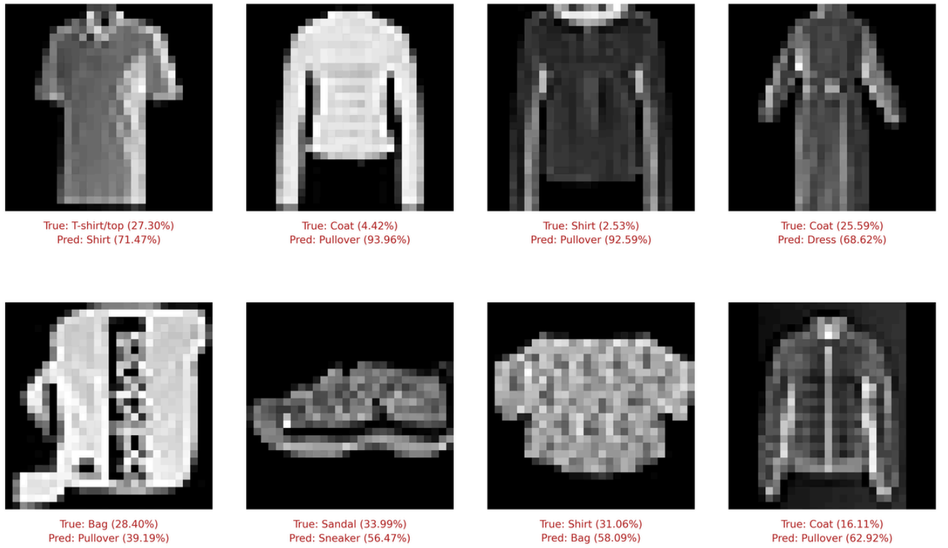
True: Pullover
Top-1 (Pred): Pullover (49.31%)
Top-2: Coat (48.73%)

Qualitative results · Transformer

Transformer / seed 42: Correct predictions



Transformer / seed 42: Incorrect predictions

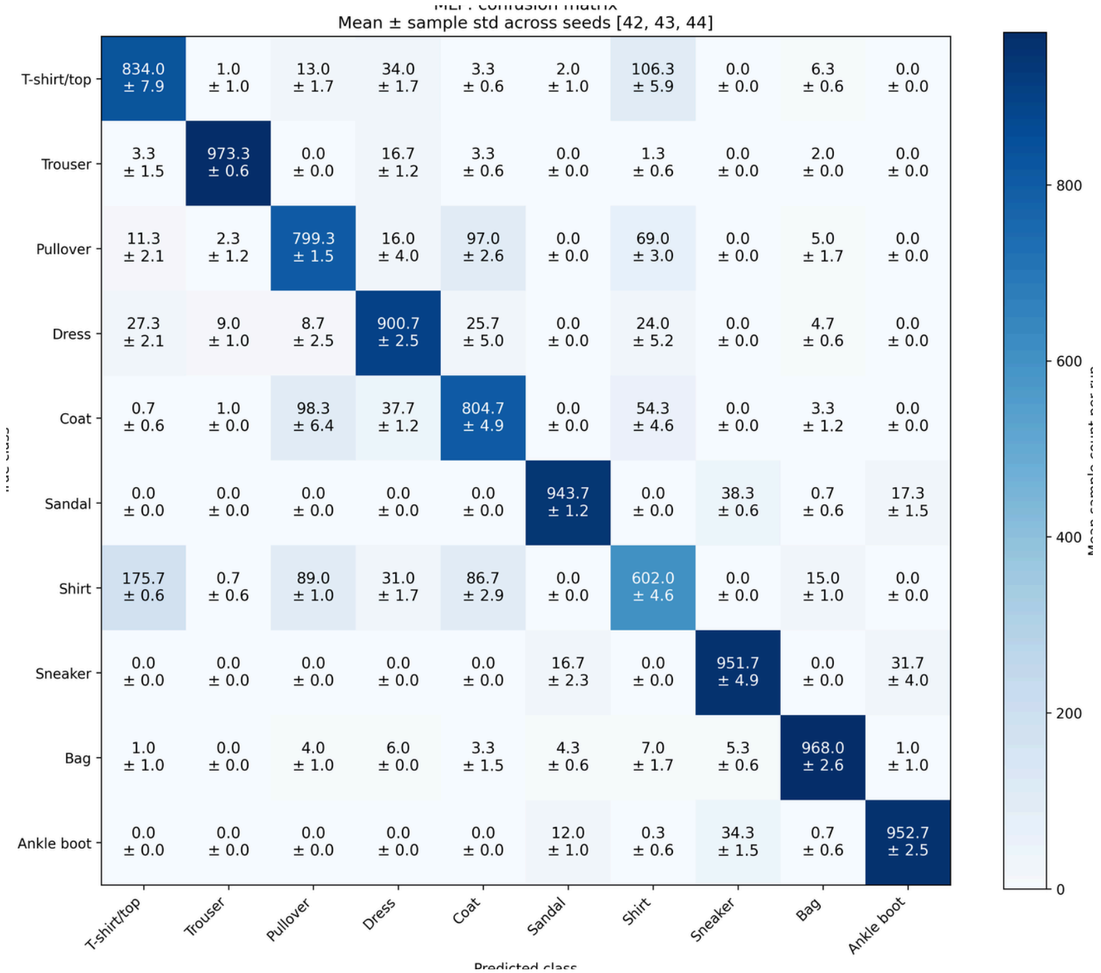


Transformer / seed 42: Difficult but correct candidates

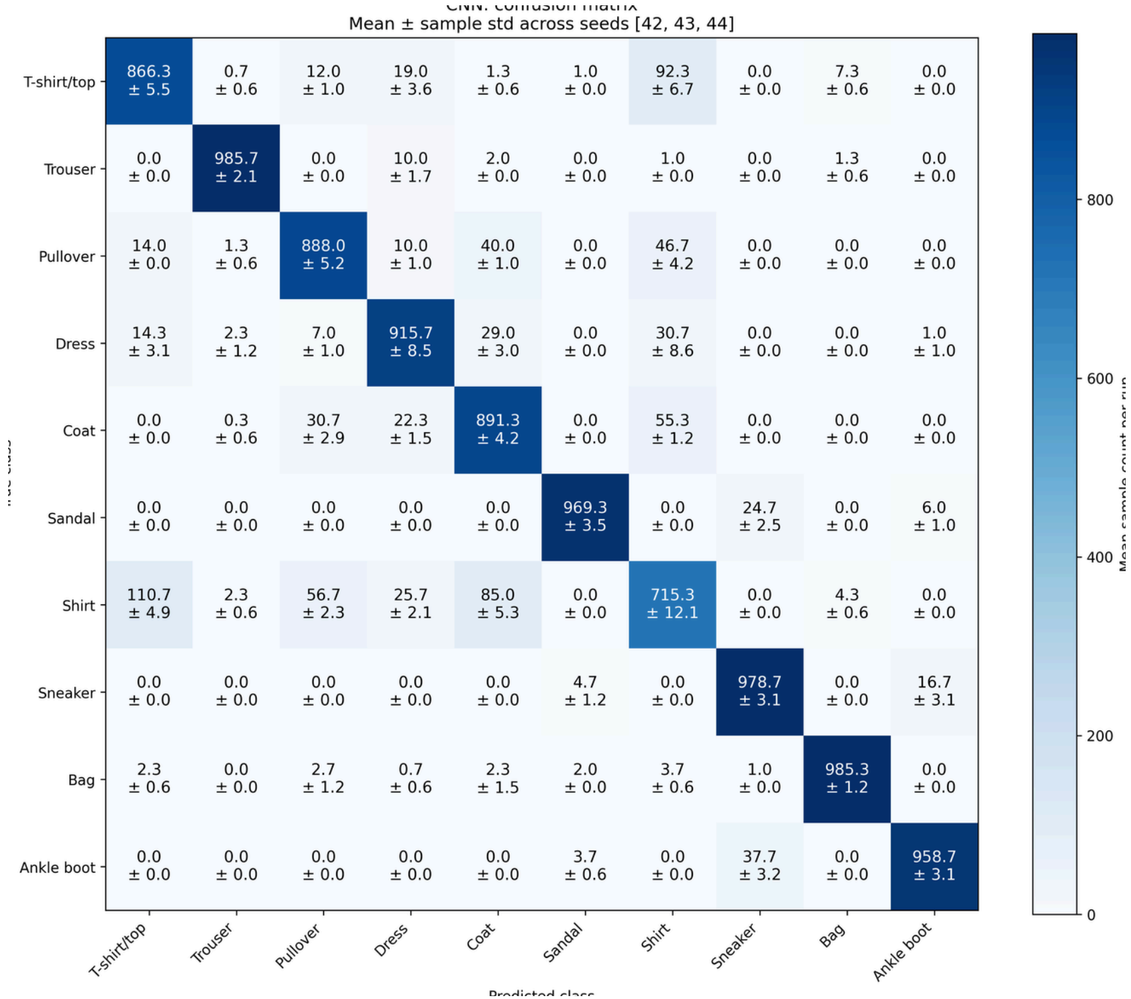


Error analysis: Confusion between visually similar classes

MLP: 1,270 mean test errors

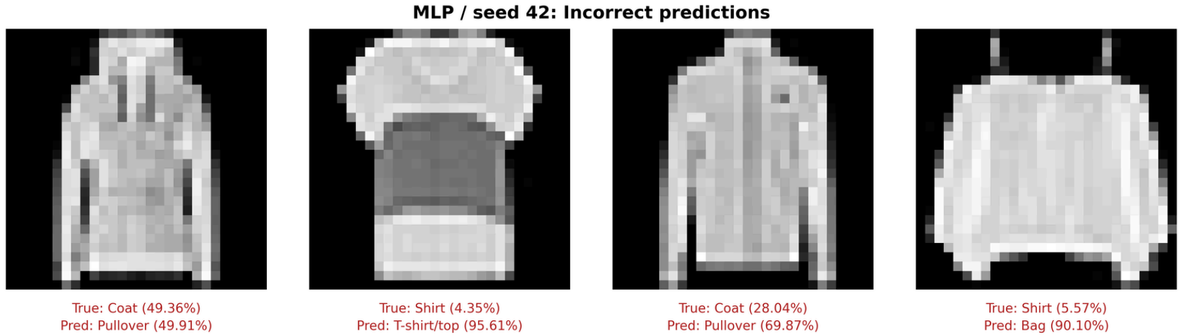


CNN: 846 mean test errors

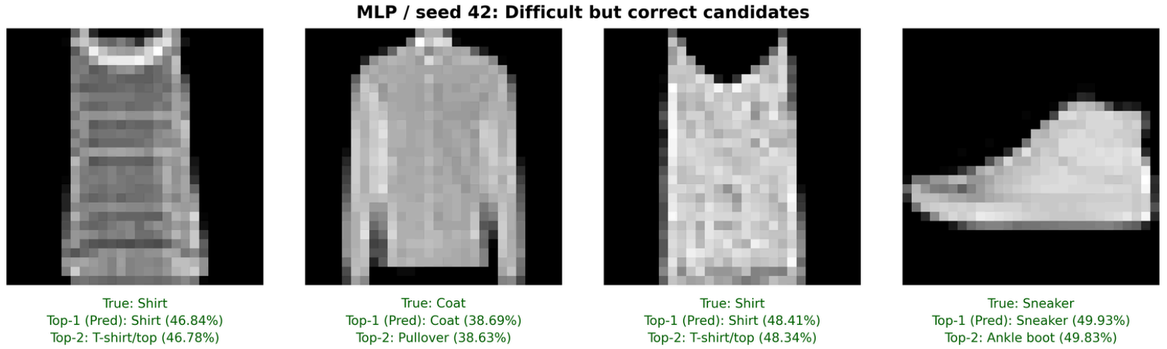


High confidence does not ensure correctness

Wrong · Shirt → Bag



Correct, but uncertain



Conclusion:

CNN leads in mean test performance

Accuracy: 91.54 ± 0.20%

Macro-F1: 91.51 ± 0.19%
231,626 parameters

All evaluated nonlinear models outperform the linear classifier on the overall predictive metrics

No model resolves all class-specific errors, and Shirt remains the most challenging category